

MATHEMATICS
Introduction to Modern Algebra I
Solutions to Midterm 2.

1) If G is a group of order p and p is a prime number, show that G is cyclic (you may use Lagrange's Theorem).

Solution: Let $H = \langle g \rangle$ be the cyclic subgroup generated by any element $e \neq g \in G$. Then as $\{e, g\} \subset H$, we have $m = |H| \geq 2$. But by Lagrange's Theorem m divides p and hence $m = p$. So the inclusion $\langle g \rangle \subset G$ is an equality.

2) If N, M are subgroups of G and N is normal in G , show that NM is a subgroup of G .

Solution: Closed under $*$: If $n, n' \in N$ and $m, m' \in M$, then $(nm)(n'm') = n(mn'm') = n(mn'm^{-1})mm' = nn''mm'$ where $n'' \in N$ and the last equality follows since N is normal in G and so $mNm^{-1} \subset N$. But as N, M are subgroups, $nn'' \in N$ and $mm' \in M$ so that $(nm)(n'm') \in NM$.

Closed under inverses: If $n \in N$ and $m \in M$ then $(nm)^{-1} = m^{-1}n^{-1} = (m^{-1}n^{-1}m)m = n''m$ where $n'' \in N$ since $m^{-1}Nm \subset N$ as N is normal in G . But then $(nm)^{-1} \in NM$.

3) Give an example of a subgroup N of a group G which is not a normal subgroup.

Solution: $N = \langle (1, 2) \rangle \subset G = S_3$. In fact $N = \{e, (1, 2)\}$ and $(1, 3)N(1, 3)^{-1} = \{(1, 3)e(1, 3), (1, 3)(1, 2)(1, 3)\} = \{e, (2, 3)\} \not\subset N$.

4) Use Lagrange's Theorem to prove Fermat's Theorem: If p is a prime number and x is an integer relatively prime to p , then x^{p-1} is congruent to 1 modulo p .

Solution: The group $(\mathbb{Z}^*, \cdot)$ has order $p-1$. Therefore the subgroup generated by $[x]$ has order m where m divides $p-1$ i.e. $p-1 = mq$. But then $[x^{p-1}] = [x]^{p-1} = ([x]^m)^q = ([1])^q = [1]$ i.e. $[x^{p-1} - 1] = 0$ or equivalently p divides $x^{p-1} - 1$. That is x^{p-1} is congruent to 1 modulo p .

5) If $G/Z(G)$ is cyclic, show that G is abelian.

Solution: Let $Z = Z(G)$. Pick $g \in G$ such that G/Z is generated by Zg . Any element in G may be written as zg^i where $z \in Z$ and $i \in \mathbb{Z}$. Given two elements of G say zg^i and $z'g^j$, we have

$$(zg^i)(z'g^j) = zz'g^i g^j = zz'g^j g^i = zg^j z'g^i = z'g^j g^i = z'g^j zg^i = (z'g^j)(zg^i)$$

where we have used the fact that $g^i g^j = g^{i+j} = g^j g^i$ and z, z' are in the center and so commute with any element of G .

6) If N is a normal subgroup of a group G , prove that G/N is a group with multiplication induced by $Ng \cdot Nh = Ngh$.

Solution: $*$ is well defined: Suppose $Ng = Ng'$ and $Nh = Nh'$ i.e. $N = Ng'g^{-1}$ and $N = Nh'h^{-1}$ so $g'g^{-1} = n \in N$ and $h'h^{-1} = m \in N$ thus $g' = ng$ and $h' = mh$ for some $n, m \in N$. Then

$$Ng'Nh' = N(g'h') = Nngmh = Nngmg^{-1}gh = Nnm'gh = Ngh$$

where we have used the fact that since N is normal, $gmg^{-1} = m' \in N$ and since $nm' \in N$, then $nm'N = N$.